

Harish Ambalam Rajasimman

[LinkedIn](#) | sreeharishar@gmail.com | 716-617-0242 | [Portfolio](#)

SUMMARY

Data Scientist and Business Intelligence Specialist with an M.S. in Data Science and **5+ years of experience** developing analytics, automation, and software solutions. Experienced in Python, SQL, statistical analysis, machine learning, and enterprise data systems. Currently supporting Allegheny County Department of Human Services through development of governance automation, metadata analytics, operational monitoring, and data-driven applications. Passionate about applying data science and software engineering to improve decision-making, service delivery, and outcomes for vulnerable populations.

SKILLS

Programming & Software Development: Python, SQL (PostgreSQL, Oracle, MS SQL Server, MySQL), R, Bash, JSON, YAML, REST API Development & Integration, Git

Machine Learning & Statistical Analysis: Predictive Modeling, Statistical Modeling, Scikit-Learn, XGBoost, Regression Modeling, Classification, Time Series Forecasting (ARIMA), A/B Testing, Hypothesis Testing, Experimental Design, PCA, NLP, Data Quality, Validation & Governance

Data Engineering & Distributed Computing: Apache Spark, Kafka, Hadoop, AWS (S3, Lambda, Redshift, SageMaker), Google Cloud (BigQuery, Vertex AI, Cloud Functions), Docker

Data Processing & Visualization: Pandas, NumPy, Plotly, Streamlit, Power BI (DAX, Power Query), Looker

Business Intelligence & Analytics Platforms: Tableau REST API, Tableau Metadata API (GraphQL), Tableau Server Client (TSC), TabPy Integration, Tableau Repository Analysis, Tableau Desktop, Tableau Prep, Tableau Server Administration

Automation & Infrastructure: Rundeck Process Automation, Linux Administration, Azure Virtual Machines, systemd Services, Logging & Monitoring, PagerDuty Integration

WORK EXPERIENCE

Business Intelligence Specialist,

Allegheny County Department of Human Services, Pittsburgh, PA

May 2025 - Present

- Develop Python-based applications and automation frameworks that support governance, compliance, operational monitoring, and analytics workflows across DHS enterprise data platforms.
- Design and maintain software integrations using Tableau Server Client (TSC), Tableau REST API, Metadata API (GraphQL), and other enterprise services to automate data collection, reporting, and administrative processes.
- Build metadata-driven analytics solutions that evaluate workbook usage, content ownership, data source lineage, dependency relationships, and engagement metrics to support governance and decision-making.
- Develop reusable software and scheduled automation workflows in Rundeck to improve operational efficiency, reduce manual effort, and standardize governance processes across Tableau Server environments.
- Create monitoring and alerting systems for enterprise analytics infrastructure, including execution monitoring, logging, error handling, and automated remediation workflows.
- Analyze operational and usage data to identify trends, optimize resource allocation, improve platform adoption, and support strategic planning initiatives.
- Collaborate with analytics, infrastructure, and security teams to implement authentication, access control, governance standards, and platform reliability improvements.
- Support enterprise Tableau Server administration including content lifecycle management, permissions, licensing, publishing standards, and platform performance optimization.

Data Analyst, Community Dreams Foundation, USA

September 2024 - May 2025

- Developed and maintained cloud-based data pipelines using AWS (S3, Lambda, Redshift) and Google Cloud (BigQuery), reducing processing time by 40% and improving data availability for analytics and reporting.
- Built machine learning models using Scikit-Learn to forecast energy consumption and identify usage patterns, improving resource allocation and operational planning by 20%.
- Conducted statistical analyses, including regression modeling and hypothesis testing, to evaluate program effectiveness and support data-driven decision-making.
- Designed data models and analytical workflows to measure project performance, environmental impact, and operational outcomes across multiple initiatives.
- Developed executive dashboards and KPI reporting solutions using Power BI and Looker, enabling stakeholders to monitor program performance and identify improvement opportunities.

- Collaborated with cross-functional teams to translate business questions into analytical solutions and actionable recommendations.
- Implemented data quality validation and governance processes to improve reporting accuracy, consistency, and reliability across datasets.
- Applied A/B testing and statistical evaluation techniques to assess program interventions and support optimization of incentive distribution strategies.

Data Analytics Specialist, SRS Electronics, India

January 2020 - January 2023

- Developed and automated business intelligence solutions using Python, SQL, Power BI, and DAX, reducing report generation time by 50% and improving access to operational insights.
- Designed and maintained cloud-based data pipelines using AWS (S3, Lambda, Redshift), improving data processing efficiency by 40% and supporting enterprise reporting workflows.
- Built predictive models using Random Forest, Gradient Boosting, and regression techniques to analyze customer behavior, improve targeting strategies, and increase marketing campaign effectiveness by 15%.
- Performed statistical analysis, hypothesis testing, and trend analysis to identify business opportunities, improve forecasting accuracy, and support strategic decision-making.
- Collaborated with business stakeholders to translate operational challenges into analytical solutions, delivering actionable insights that improved sales performance and reporting effectiveness.
- Implemented data quality validation processes and reporting standards to improve accuracy, consistency, and reliability across business intelligence initiatives.

SELECTED TECHNICAL PROJECTS

Tableau Governance Automation Framework | Python, Tableau REST API, Tableau Metadata API, Tableau Server Client (TSC), Rundeck

- Developed a suite of automated governance solutions that identify ownership, permission, lifecycle, and content management issues across enterprise Tableau Server environments.
- Built reusable Python applications integrating Tableau REST API, Metadata API, and Tableau Server Client libraries to automate auditing, reporting, and remediation workflows.
- Reduced manual governance effort through scheduled automation and metadata-driven monitoring processes.

Metadata Analytics & Lineage Reporting Platform | Python, GraphQL, Tableau Metadata API, SQL

- Designed metadata analytics solutions to analyze workbook dependencies, data source lineage, content ownership, extract utilization, and user engagement across Tableau Server.
- Developed automated reporting workflows that provide visibility into content relationships and support governance decision-making.
- Enabled stakeholders to identify optimization opportunities, improve content management, and support platform-wide governance initiatives.

Enterprise Monitoring & Alerting Framework | Python, Rundeck, REST APIs, Linux

- Developed automated monitoring solutions for enterprise analytics infrastructure, including job execution monitoring, logging, alerting, and failure detection.
- Implemented automated remediation and notification workflows to improve operational reliability and reduce manual intervention.
- Built scheduled monitoring processes capable of identifying long-running executions, operational failures, and infrastructure issues across analytics environments.

EDUCATION

University at Buffalo, The State University of New York

Master of Science in Data Science

June 2024

Relevant Coursework: Machine Learning, Statistical Analysis, Data Intensive Computing, Data Visualization, Applied Data Mining